The Six-Dimensional Case of the Skew-Hermitian Lie Algebra Problem

Automatic Math Discovery Smoke Test

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Source problem

The source paper asks for the possible dimensions of the Lie algebra $\mathcal{Z}(X)$ of skew-hermitian operators when X is a finite-dimensional real Banach space. It notes that dimensions 3 and 4 were known and that the first open case was dimension 5.

> Our next aim is to discuss some questions related to the Lie algebra of skew-hermitian operators $\mathcal{Z}(X)$ of an arbitrary n-dimensional space which. The main related open question is the following.
>
> **Problem 27.** *Figure out what are the possible values for the dimension of* $\mathcal{Z}(X)$ *when* $\dim(X) = n$.
>
> Let us fix a n-dimensional Banach space X. It follows from a theorem of Auerbach [72, Theorem 9.5.1], that there exists an inner product $(\cdot|\cdot)$ on X such that every skew-hermitian operator on X remains skew-hermitian (hence skew-symmetric) on $H := (X, (\cdot|\cdot))$. Then, by just fixing an orthonormal basis of H, we get an identification of $\mathcal{Z}(X)$ with a Lie subalgebra of the Lie algebra $\mathcal{A}(n)$. Therefore,
> $$\dim\big(\mathcal{Z}(X)\big) \leqslant \frac{n(n-1)}{2}.$$
> The equality holds if and only if X is a Hilbert space (see [73, Theorem 3.2] or [60, Corollary 2.7]). It is a good question whether or not all the intermediate numbers are possible values for the dimension of $\mathcal{Z}(X)$. The answer is negative, as a consequence of Theorem 3.2 in Rosenthal's paper [73], which reads as follows.
>
> (a) If $\dim\big(Z(X)\big) > \frac{(n-1)(n-2)}{2}$, then X is a Hilbert space and so $\dim\big(Z(X)\big) = \frac{n(n-1)}{2}$.
>
> (b) $\dim\big(Z(X)\big) = \frac{(n-1)(n-2)}{2}$ if and only if X is a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$.
>
> For low dimensions, Problem 27 has been solved in [74]. When the dimension of X is 3, the above result leaves only the following possible values for the dimension of $\mathcal{Z}(X)$: 0 as for $X = \ell_\infty^3$, 1 as for $\mathbb{R} \oplus_1 \mathbb{C}$, and 3 as for ℓ_2^3. When the dimension of X is 4, the possible values of the dimension of $\mathcal{Z}(X)$ allowed by the above result are $0, 1, 2, 3, 6$; all of them are possible [74, pp. 443]. The first dimension in which Problem 27 is open is $n = 5$.
>
> **Problem 28.** *What are the possible values for the dimension of* $\mathcal{Z}(X)$ *when* X *is a* 5*-dimensional real Banach space?*

Result

Theorem 1. *Let* X *be a six-dimensional real Banach space. Then*

$$\dim \mathcal{Z}(X) \in \{0, 1, 2, 3, 4, 6, 7, 10, 15\}.$$

Conversely, every value in this set is attained by some six-dimensional real Banach space.

Proof. By the Auerbach reduction quoted in the source paper, $\mathcal{Z}(X)$ embeds as a compact Lie subalgebra of $\mathfrak{so}(6)$. Rosenthal's gap theorem says that, for $\dim X = n$, if

$$\dim \mathcal{Z}(X) > \frac{(n-1)(n-2)}{2},$$

then X is Hilbert and $\dim \mathcal{Z}(X) = n(n-1)/2$; and equality at $(n-1)(n-2)/2$ occurs exactly for a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$. For $n = 6$, this leaves only the values

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15.$$

The value 10 is the equality case in Rosenthal's theorem. It remains to exclude 5, 8, and 9.

We use two elementary compact Lie algebra facts. First, a compact Lie algebra is the direct sum of its center and its compact semisimple part. Second, the only compact simple Lie algebras of dimension at most 9 are $\mathfrak{su}(2)$, dimension 3, and $\mathfrak{su}(3)$, dimension 8.

Excluding dimension 5**.** A five-dimensional compact Lie algebra in $\mathfrak{so}(6)$ cannot be abelian, since an abelian subalgebra of $\mathfrak{so}(6)$ has dimension at most $\operatorname{rank} \mathfrak{so}(6) = 3$. Thus it must be isomorphic to $\mathfrak{su}(2) \oplus \mathbb{R}^2$.

Consider the associated faithful orthogonal representation on $\mathbb{R}^6$. The nontrivial real irreducible $\mathfrak{su}(2)$-modules of dimension at most 6 have dimensions 3, 4, and 5. The central $\mathbb{R}^2$ must act through the skew-symmetric commutant of the $\mathfrak{su}(2)$ action. A case check gives only one way for this commutant to have rank at least 2: the $\mathfrak{su}(2)$ module is a four-dimensional quaternionic irreducible plus two trivial dimensions. In that case the relevant commutant is $\mathfrak{sp}(1) \oplus \mathfrak{so}(2)$, and the faithful $\mathbb{R}^2$ uses a circle in the $\mathfrak{sp}(1)$ factor and the $\mathfrak{so}(2)$ on the trivial two-plane.

Hence the connected group generated on the four-dimensional summand is $\mathrm{SU}(2)$ together with one commuting circle, i.e. $\mathrm{U}(2)$ on $\mathbb{C}^2$. This group is transitive on the Euclidean sphere of that four-dimensional summand. If a norm on $\mathbb{R}^4 \oplus \mathbb{R}^2$ is invariant under this action and under the rotation of the two-plane, then for each fixed two-plane component it depends on the first component only through its Euclidean length, and for each fixed first component it depends on the two-plane component only through its Euclidean length. Therefore the norm is invariant under $\mathrm{SO}(4) \times \mathrm{SO}(2)$, whose Lie algebra has dimension $6 + 1 = 7$. Thus a six-dimensional Banach space cannot have $\dim \mathcal{Z}(X) = 5$.

Excluding dimension 8**.** Let $\mathfrak{g} \subset \mathfrak{so}(6)$ be compact of dimension 8. If the center of $\mathfrak{g}$ had positive dimension large enough to make the semisimple part 0 or $\mathfrak{su}(2)$, it would contain an abelian subalgebra of dimension at least 5, impossible in $\mathfrak{so}(6)$. If the semisimple part were $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, the center would have dimension 2; but the faithful six-dimensional orthogonal representations of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ are, up to adding trivial summands, either the $3+3$ adjoint representation or the four-dimensional real spin representation of $\mathfrak{so}(4)$ plus a two-dimensional trivial summand. Their skew-symmetric commutants have rank 0 and 1, respectively, so a two-dimensional center cannot act faithfully.

The remaining possibility is $\mathfrak{g} \cong \mathfrak{su}(3)$ acting on $\mathbb{C}^3$ viewed as $\mathbb{R}^6$. The group $\mathrm{SU}(3)$ is transitive on the unit sphere of $\mathbb{C}^3$. Therefore every $\mathrm{SU}(3)$-invariant norm on $\mathbb{R}^6$ is Euclidean, and its skew-hermitian Lie algebra is all of $\mathfrak{so}(6)$, of dimension 15. Thus dimension 8 is impossible.

Excluding dimension 9**.** Let $\mathfrak{g} \subset \mathfrak{so}(6)$ be compact of dimension 9. The possible semisimple parts are 0, $\mathfrak{su}(2)$, $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\mathfrak{su}(3)$, and $\mathfrak{su}(2)^3$.

The first two require a center of dimension at least 6, impossible because abelian subalgebras of $\mathfrak{so}(6)$ have dimension at most 3. The case $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ would require a three-dimensional center, but the centralizer ranks in all faithful six-dimensional orthogonal representations listed above are at most 1.

For the $\mathfrak{su}(2)^3$ case, no faithful real orthogonal representation has dimension 6: a nontrivial real irreducible of one $\mathfrak{su}(2)$ has dimension at least 3, the smallest faithful module for two factors has real dimension 4, and the smallest way to include the third factor already has dimension at least $4 + 3 = 7$.

Thus the only remaining case is $\mathfrak{su}(3) \oplus \mathbb{R}$, acting as $\mathfrak{u}(3)$ on $\mathbb{C}^3$. The group U(3) is transitive on the unit sphere of $\mathbb{C}^3$, so again any invariant norm is Euclidean and the skew-hermitian Lie algebra enlarges to $\mathfrak{so}(6)$, dimension 15. Therefore dimension 9 is impossible.

We now realize the remaining values. Let H_k denote the real Hilbert space of dimension k. For a finite ℓ_∞-sum of Euclidean blocks, the identity component of the isometry group is the product of the orthogonal groups on the Euclidean blocks, so the Lie algebra dimension is $\sum_j k_j(k_j - 1)/2$. Thus the following six-dimensional spaces realize the announced values:

$\dim \mathcal{Z}(X)$	X
0	ℓ_∞^6
1	$H_2 \oplus_\infty \ell_\infty^4$
2	$H_2 \oplus_\infty H_2 \oplus_\infty \ell_\infty^2$
3	$H_3 \oplus_\infty \ell_\infty^3$
4	$H_3 \oplus_\infty H_2 \oplus_\infty \mathbb{R}$
6	$H_4 \oplus_\infty \ell_\infty^2$
7	$H_4 \oplus_\infty H_2$
10	$H_5 \oplus_\infty \mathbb{R}$
15	H_6.

This proves both necessity and realization. □

Verification notes

The script `code/check_six_dimension_values.py` lists the Euclidean-block construction values and confirms that Rosenthal's theorem reduces the finite exclusion list to 5, 8, and 9. It is only a sanity check; the proof of the exclusions is the compact Lie algebra argument above.

Novelty check

Before packaging this candidate partial result, I searched the local parsed corpus and run artifacts for the exact source arXiv id and phrases such as `dim Z(X)`, `dimension of mathcal-Z(X)`, `six-dimensional real Banach`, and `skew-hermitian`. The only nearby results found in this run were the source problem statement and the five-dimensional packet.

I also ran bounded web searches on 2026-06-13 using exact and near-exact phrases around "dim Z(X)", "six-dimensional real Banach space", "skew-hermitian operators", and Rosenthal's Lie algebra paper. These searches found adjacent numerical-index and skew-hermitian-operator literature, but no direct later classification of the six-dimensional case.

Scope

This is a candidate solution of the six-dimensional subcase of the general possible-values problem. It does not solve the arbitrary-dimensional problem.